AMSE: all available evidence

Finding

Full AMSE improves spectral-power fidelity but sacrifices forecast accuracy. Lower AMSE weights retain more accuracy, yet no candidate in the completed joint campaign meets all three aggregate requirements plus the safety guard.

Coverage

September 19 full-AMSE/MSE pair; September 24 lambda=0.1 blend; October 4 trajectory re-evaluation, 500-update pilot, stopped/differentiable feedback, and three objective ablations. All 49 evaluated records across the four October phases are included, with repeated controls retained separately.

What is still pending

All six 2,000-update confirmation training runs finished. Seven evaluation tasks 15149394_0–6 and report 15149395 remain pending at the live queue check on October 7. Confirmation covers three seeds and 64 separate seasonal 2023 starts. No confirmation scores are available.

Related MGDA experiment

MGDA combines GraphCast and spectral-power gradients. Calibration completed; both 100-update smoke runs failed bitwise resume equality of residual parameters. The 2,000-update production comparison did not launch. These are engineering results, not evidence of forecast skill.

Reading this report

Positive reduction means better than Original GraphCast. Exact GraphCast loss is the headline metric; physical RMSE and relative spectral-power error are separate diagnostics. The companion complete PDF appends the existing 49-page field-level spectral report.

Protocols and interpretation

Historical accuracy versus spectral diagnostics

Historical exact-loss evaluation uses 32 matched 2022 starts and 40 six-hour leads. Spectral evaluation uses eight previously inspected 2023 starts, 30–60° latitude in both hemispheres, and zonal wavenumbers 1–180. Neither is the pending year-round 2023 evaluation.

Objective and measurements

Training AMSE interpolates spherical spectral MSE (lambda=0) and full AMSE (lambda=1), using a global T179 transform. Verification uses zonal Fourier spectra. Exact GraphCast loss retains its original normalization, latitude, pressure and variable weights; losses are pooled before the reduction ratio.

October joint qualification

On the same eight spectral starts: exact GC loss ratio <1, mean physical-RMSE ratio <1, and mean relative-power-error ratio <1; additionally no selected primitive field/day may exceed 2% RMSE harm versus GC. RMSE/power summaries use nine fields and days 1, 3, 7, 10. Power error is averaged over spectral bins then field/day cells.

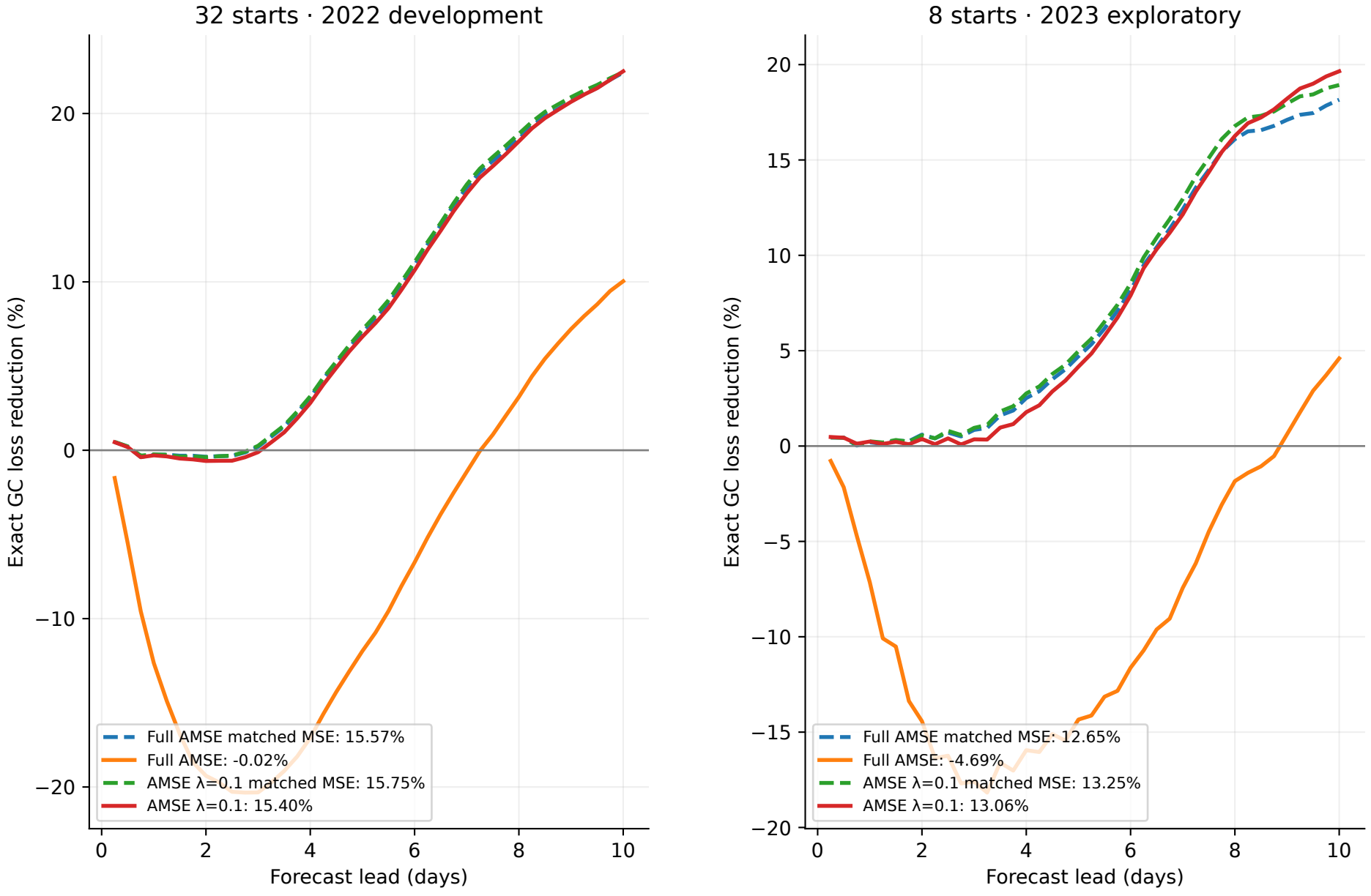
Baseline and uncertainty limits

Historical comparisons flag numerical baseline drift and use each model’s paired GC baseline. October scores use their phase’s canonical GC reference; controls vary between re-evaluations. Do not average repeated baselines or interpret tiny differences as established gains. Development selection has no confirmatory uncertainty claim.

Power is not accuracy

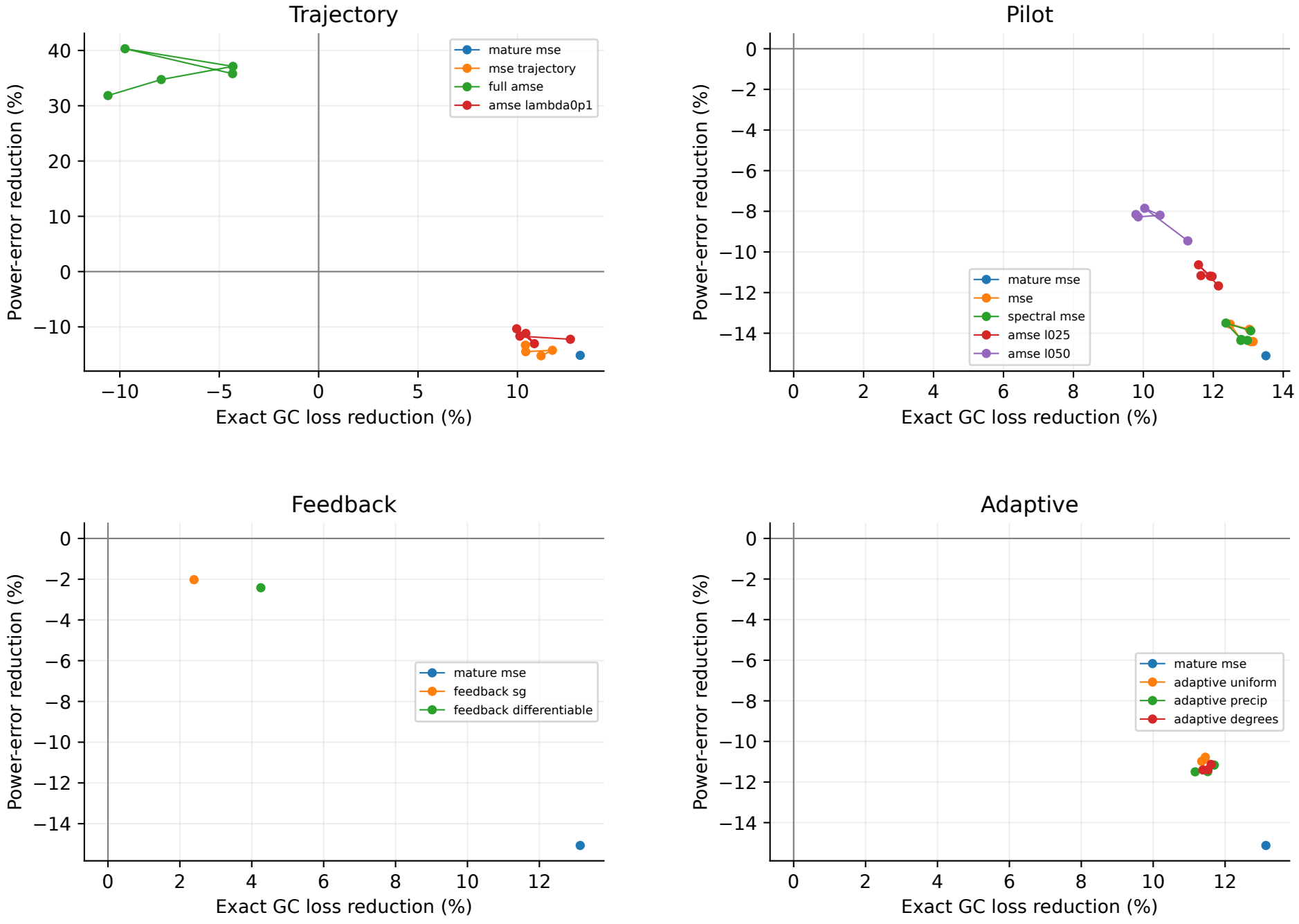
Matching power does not ensure the correct location or phase of weather features. Coherence and spectral-error decomposition in the appendix expose this distinction. No new forecast inference or training was run to prepare this report.

Historical AMSE: accuracy by forecast lead



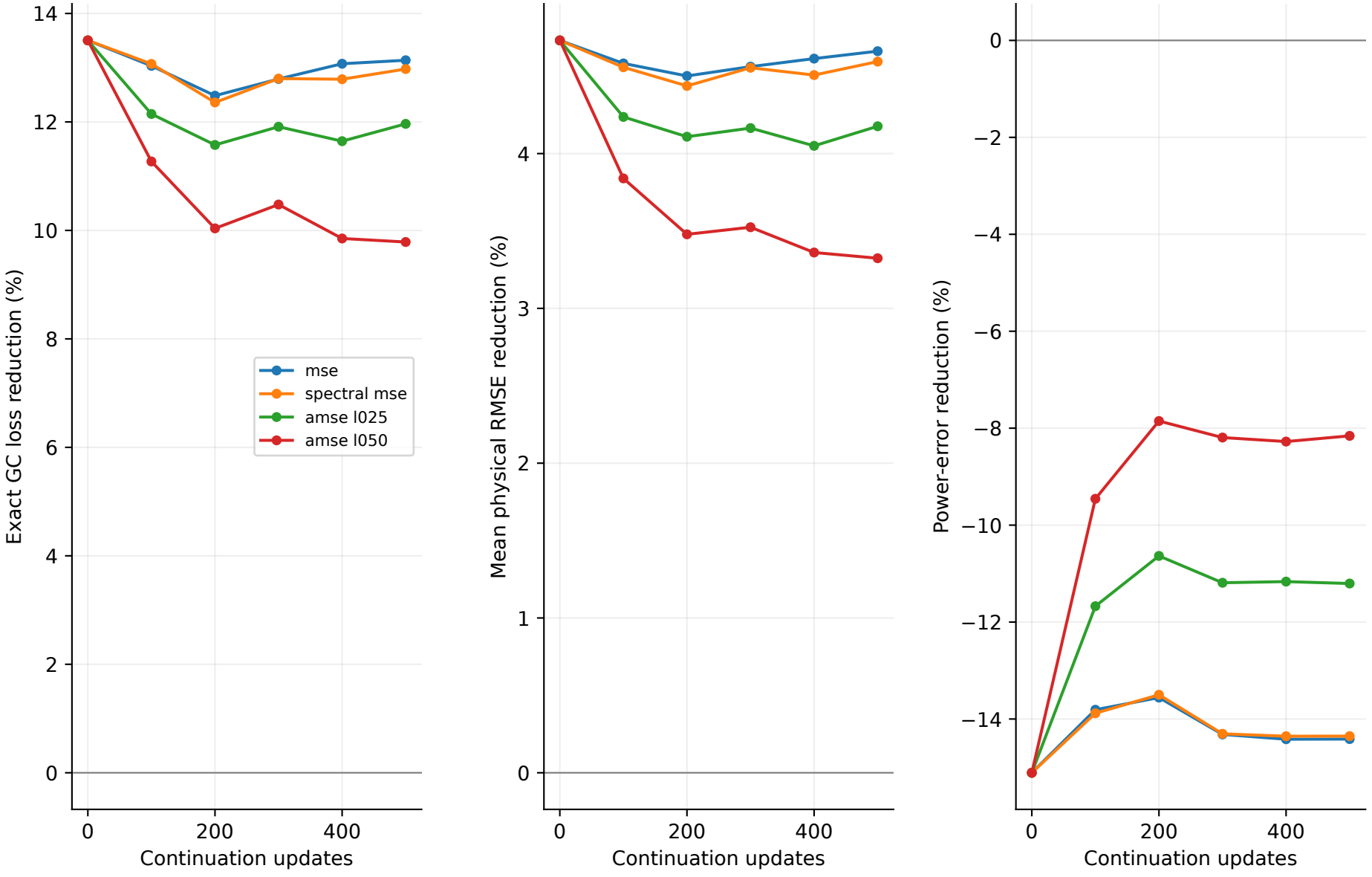
SWA 500–2000; paired baselines retained separately. Legend gives pooled rollout reduction, not mean lead percentage.

Accuracy and power: no joint winner yet



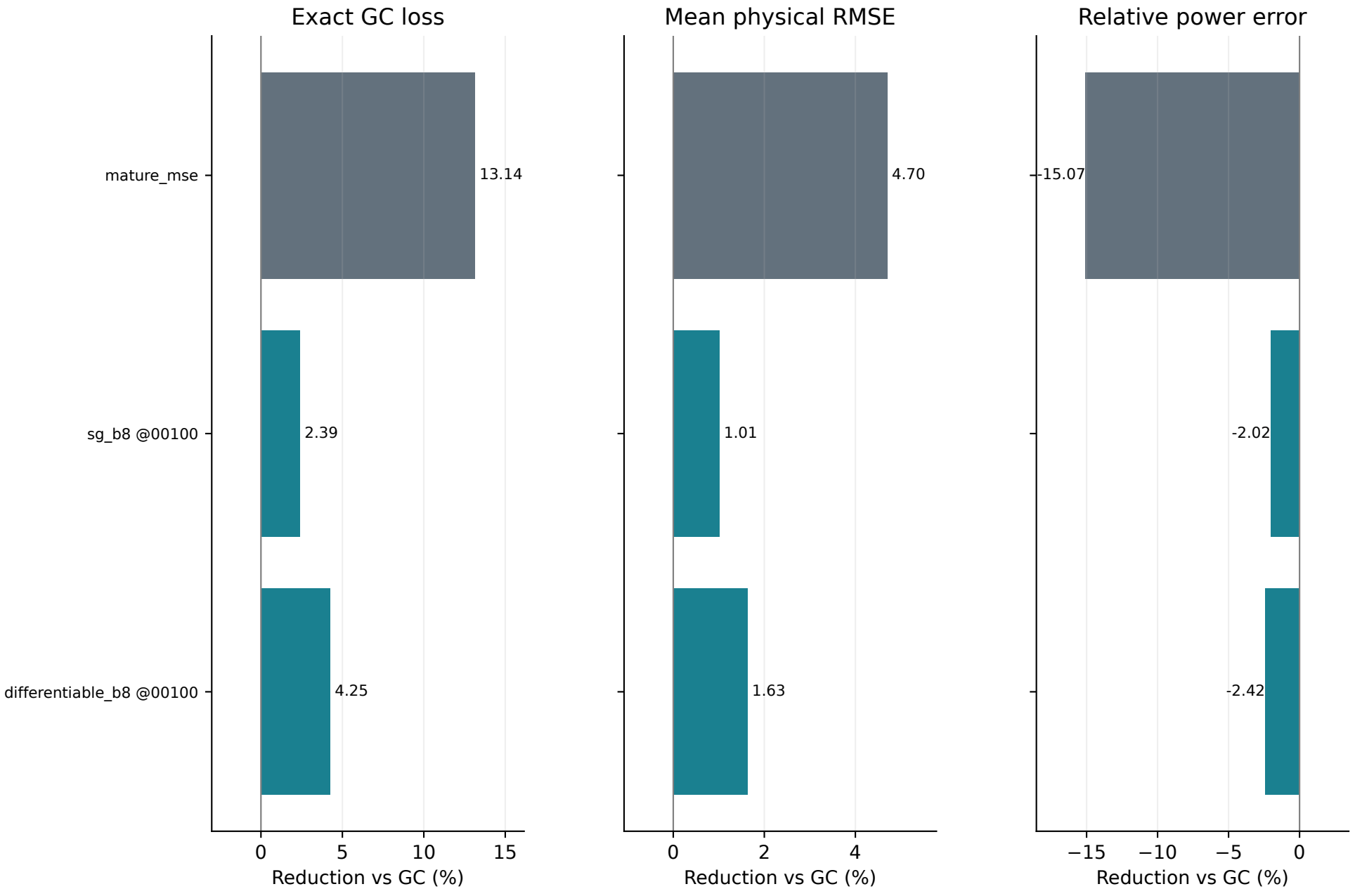
Same eight 2023 starts. Upper right improves both axes; qualification additionally requires physical RMSE and safety.

500-update pilot: increasing AMSE trades accuracy for power



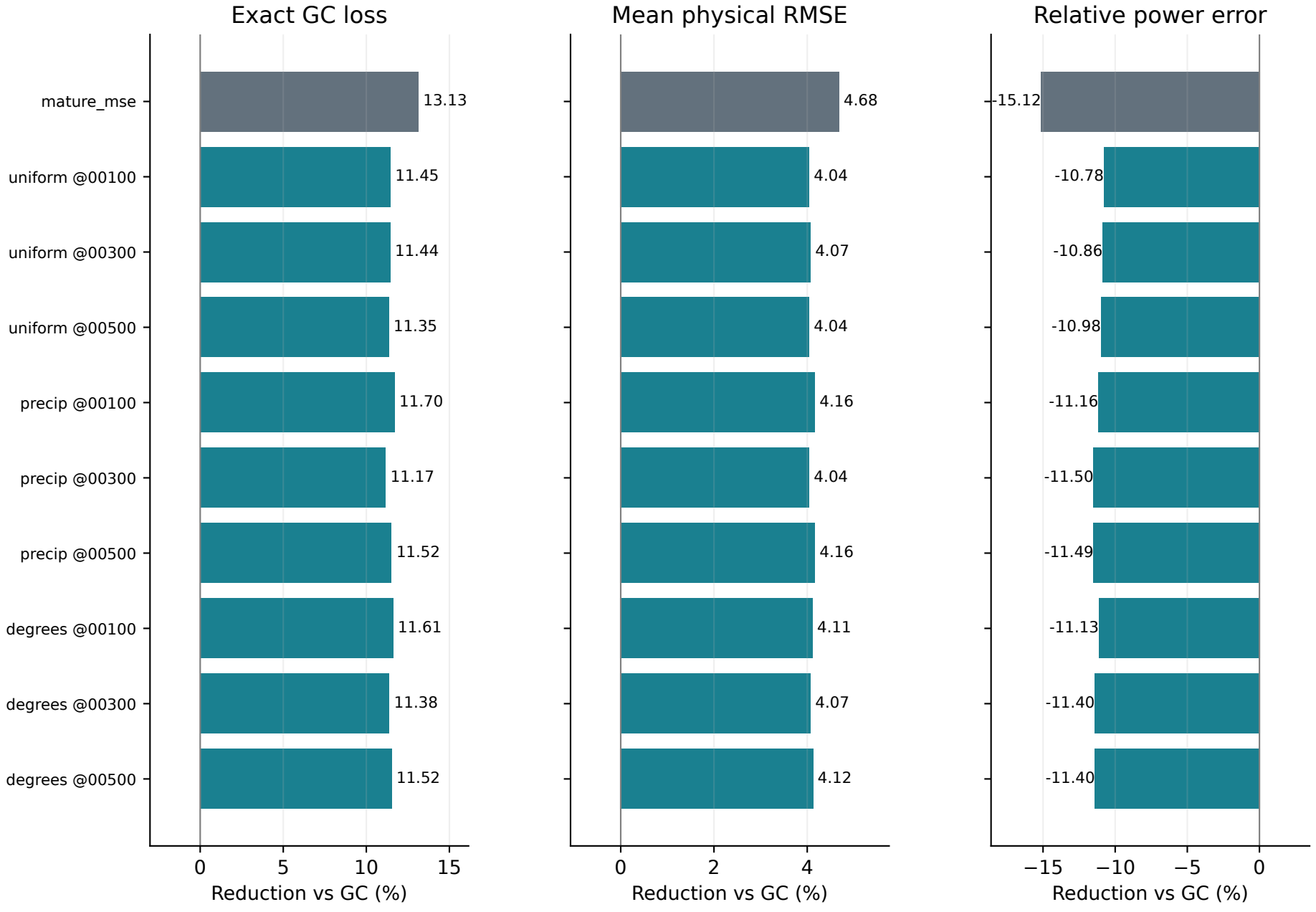
Common mature-MSE start; four matched 500-update continuations. All checkpoints shown; all fail joint qualification.

Feedback experiments: completed development results



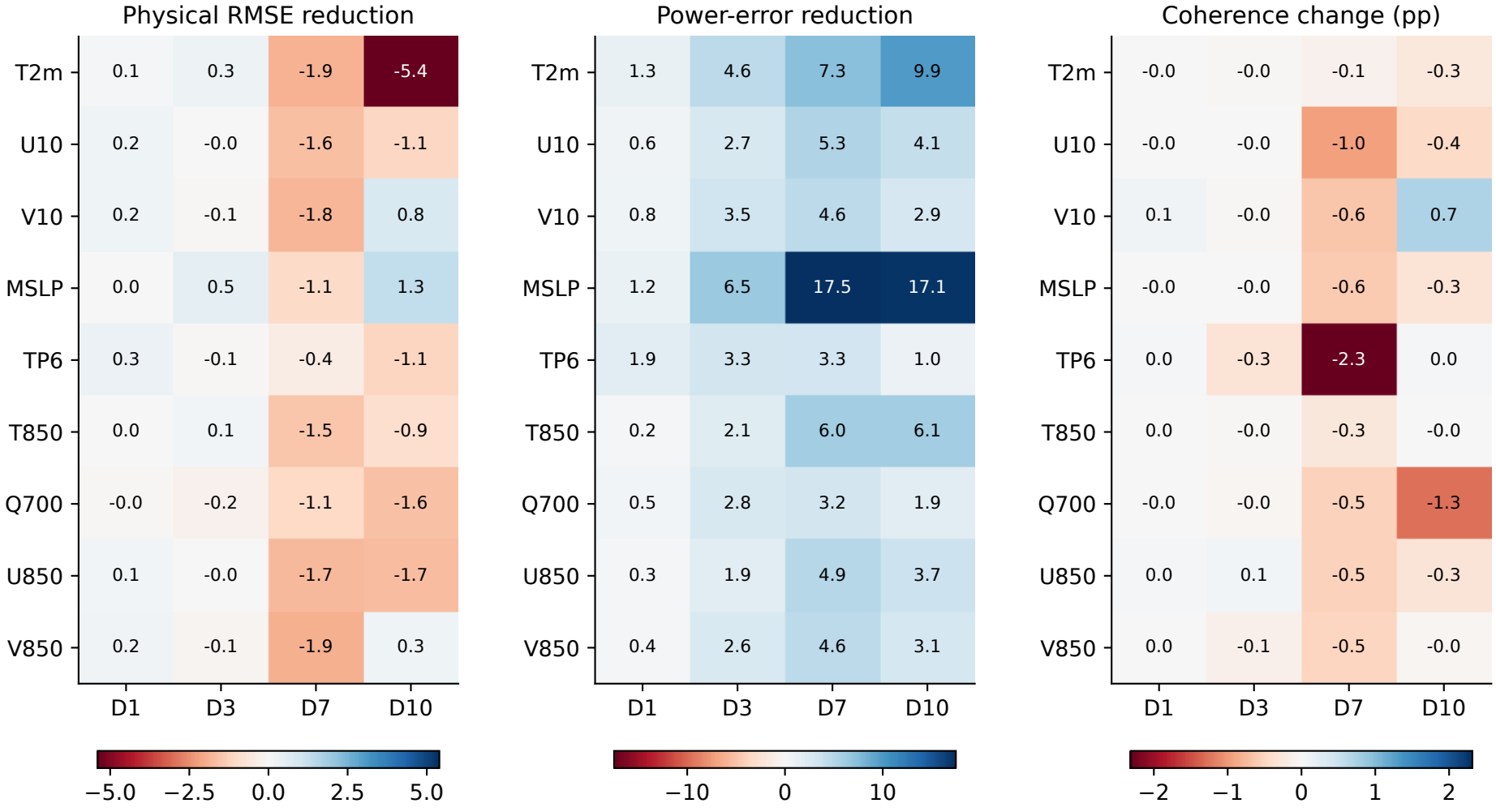
Exploratory eight-start 2023 diagnostics. Positive is better. No candidate passes the joint qualification gate.

Adaptive experiments: completed development results



Exploratory eight-start 2023 diagnostics. Positive is better. No candidate passes the joint qualification gate.

Locked recipe: where does it help or hurt?



adaptive_precip @100 versus phase MSE control. Positive is better; blue helps, red harms. This is development, not confirmation.

Complete checkpoint inventory: trajectory

Candidate	GC loss %	RMSE %	Power error %	Joint cells	Safety	Eligible
mature_mse	13.16	4.64	-15.15	8/36	pass	no
mse_trajectory_step00500	10.40	3.57	-13.31	5/36	pass	no
mse_trajectory_step01000	10.42	3.74	-14.49	5/36	pass	no
mse_trajectory_step01500	11.76	4.04	-14.23	9/36	pass	no
mse_trajectory_step02000	11.19	4.12	-15.22	9/36	pass	no
full_amse_step00500	-10.60	-5.28	31.84	2/36	FAIL	no
full_amse_step01000	-7.92	-3.90	34.74	4/36	FAIL	no
full_amse_step01500	-4.31	-2.63	37.12	8/36	FAIL	no
full_amse_step02000	-9.74	-4.46	40.31	1/36	FAIL	no
full_amse_swa	-4.33	-2.73	35.80	8/36	FAIL	no
amse_lambda0p1_step00500	9.96	3.30	-10.35	6/36	pass	no
amse_lambda0p1_step01000	10.85	3.71	-13.04	3/36	pass	no
amse_lambda0p1_step01500	10.42	3.56	-11.19	9/36	pass	no
amse_lambda0p1_step02000	10.12	3.69	-11.68	11/36	pass	no
amse_lambda0p1_swa	12.66	4.35	-12.24	9/36	pass	no

All percentage columns are reductions versus GC; positive is better. Joint cells improve both physical RMSE and power error.

Complete checkpoint inventory: pilot

Candidate	GC loss %	RMSE %	Power error %	Joint cells	Safety	Eligible
mature_mse	13.50	4.73	-15.11	9/36	pass	no
mse_step00100	13.03	4.58	-13.81	11/36	pass	no
mse_step00200	12.48	4.50	-13.56	11/36	pass	no
mse_step00300	12.79	4.56	-14.32	12/36	pass	no
mse_step00400	13.07	4.61	-14.42	11/36	pass	no
mse_step00500	13.14	4.66	-14.41	11/36	pass	no
spectral_mse_step00100	13.07	4.56	-13.88	11/36	pass	no
spectral_mse_step00200	12.36	4.44	-13.50	11/36	pass	no
spectral_mse_step00300	12.80	4.55	-14.31	11/36	pass	no
spectral_mse_step00400	12.79	4.51	-14.35	10/36	pass	no
spectral_mse_step00500	12.98	4.59	-14.35	11/36	pass	no
amse_l025_step00100	12.15	4.24	-11.67	13/36	pass	no
amse_l025_step00200	11.58	4.11	-10.64	13/36	pass	no
amse_l025_step00300	11.91	4.16	-11.19	13/36	pass	no
amse_l025_step00400	11.65	4.05	-11.17	12/36	pass	no
amse_l025_step00500	11.96	4.18	-11.20	12/36	pass	no
amse_l050_step00100	11.27	3.84	-9.46	11/36	pass	no
amse_l050_step00200	10.04	3.48	-7.85	10/36	pass	no
amse_l050_step00300	10.48	3.52	-8.19	8/36	pass	no
amse_l050_step00400	9.85	3.36	-8.28	7/36	pass	no
amse_l050_step00500	9.79	3.32	-8.16	8/36	pass	no

All percentage columns are reductions versus GC; positive is better. Joint cells improve both physical RMSE and power error.

Complete checkpoint inventory: feedback

Candidate	GC loss %	RMSE %	Power error %	Joint cells	Safety	Eligible
mature_mse	13.14	4.70	-15.07	10/36	pass	no
feedback_sg_b8_step00100	2.39	1.01	-2.02	8/36	pass	no
feedback_differentiable_b8_step00100	4.25	1.63	-2.42	13/36	pass	no

All percentage columns are reductions versus GC; positive is better. Joint cells improve both physical RMSE and power error.

Complete checkpoint inventory: adaptive

Candidate	GC loss %	RMSE %	Power error %	Joint cells	Safety	Eligible
mature_mse	13.13	4.68	-15.12	9/36	pass	no
adaptive_uniform_step00100	11.45	4.04	-10.78	13/36	pass	no
adaptive_uniform_step00300	11.44	4.07	-10.86	12/36	pass	no
adaptive_uniform_step00500	11.35	4.04	-10.98	12/36	pass	no
adaptive_precip_step00100	11.70	4.16	-11.16	15/36	pass	no
adaptive_precip_step00300	11.17	4.04	-11.50	12/36	pass	no
adaptive_precip_step00500	11.52	4.16	-11.49	11/36	pass	no
adaptive_degrees_step00100	11.61	4.11	-11.13	12/36	pass	no
adaptive_degrees_step00300	11.38	4.07	-11.40	12/36	pass	no
adaptive_degrees_step00500	11.52	4.12	-11.40	12/36	pass	no

All percentage columns are reductions versus GC; positive is better. Joint cells improve both physical RMSE and power error.

Pending work and what the evidence supports

Confirmation: training complete, evaluation pending

Six final step2000 checkpoints exist: MSE and locked AMSE recipe at seeds 22, 23 and 24. Evaluation uses 64 seasonal 2023 starts and averages separate-model metrics, never forecast fields. The recipe is lambda=0.25 with precipitation lambda=0; it was not qualified on development. Selection is locked before confirmation.

MGDA: blocked at engineering validation

Calibration job 15069232 completed. Smoke jobs 15069233_0 and _1 reached 100 updates but failed uninterrupted-versus-resumed bitwise residual-parameter equality. A failure shared by MGDA and MSE does not demonstrate failure of the scientific objective. No production forecast comparison is available.

2023 scope

The eight-case spectral diagnostics have completed and were inspected during development. The separate 122-start year-round memory/accuracy campaign remains pending numerical qualification; the earlier daily campaign failed baseline agreement. The 64-start AMSE confirmation is a third, separate evaluation.

Supported conclusion

Full AMSE produces materially better power fidelity with worse exact forecast loss. Weak AMSE produces an intermediate trade-off. The 500-update continuations, feedback tests and adaptive variants have not established a joint improvement. Small differences are particularly sensitive to observed numerical drift.

Recommended next evidence

Finish the already queued confirmation evaluations and inspect all fields, all seeds and paired intervals.
Repair and qualify resume/numerical behavior before running MGDA production or releasing year-round evaluation.
Preserve negative outcomes and keep selection fixed.